

Tinkering & Mentoring - VACC 201



Final Project Presentation & Demonstration



Natural And Biosynthesis Of Azadirachtin

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Team No.: 07

Semester: 2nd

Program: B.Sc. Biotechnology

School: School of Science

Student Team

Name	Enrollment No.	Key Responsibility
Neel Mehta	25sc02037	Article Review & Research
Palash Panwar	25sc02039	Presenter
Harshit Parmar	25sc02040	Experimental Designer(Leader)
Karina Parmar	25sc02041	Experimental Designer
Keya Parmar	25sc02042	Article Review & Research
Krishna Parmar	25sc02043	Presentation making

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Faculty Mentor

Name	Designation	Department, School
Dr. Swati Mohapatra	Asst. Professor	Department of Life Sciences, SOS

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Introduction of Tinkering Project

Title of Tinkering Project: Natural And Biosynthesis Of Azadirachtin

Sector: Life Science

Brief: This project focuses on the use of neem (*Azadirachta indica*) for the production of azadirachtin, a natural biopesticide. It is prepared from the neem leaves extract that provides sustainable and eco-friendly alternative to chemical pesticides.

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Methodology and Tinkering Process

Project Approach- Neem leaves were extracted using Soxhlet method, concentrated using rotary evaporator, and tested for antimicrobial activity.

The Tinkering Journey- After finalising the idea some adjustments were made in extractions, solvent quantity and concentrations. Small errors like contamination and improper loading were corrected during repeated experiments.

Role of Mentoring in the Process- Guidance of seniors and lab assistants was provided during handling the laboratory instruments

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Tools and Technology Used

Method for neem extraction : Soxhlet Method

To separate solvent: Rotary Vacuum Evaporator

Testing: Microbial culture in laminar air flow

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Project Outcome & Development

Procedure:

Preparation of Plant Material

- Fresh neem leaves were collected
- Leaves were washed with distilled water
- Air-dried at room temperature
- Dried leaves were ground into fine powder

Soxhlet Extraction

- 20–30 g of powdered leaves placed in a thimble
- Extraction carried out using a solvent (e.g., methanol)
- Heated in Soxhlet apparatus for 6–8 hours
- Continuous extraction cycles performed until solvent became colorless

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Project Outcome & Development

Procedure:

Concentration of Extract

- Extracted solution collected
- Solvent was removed using rotary vacuum evaporator
- Concentrated extract obtained

Preparation of Test Sample

- Extract dissolved in small amount of solvent
- Required concentrations prepared for testing

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Project Outcome & Development

Procedure:

Antimicrobial Activity Test

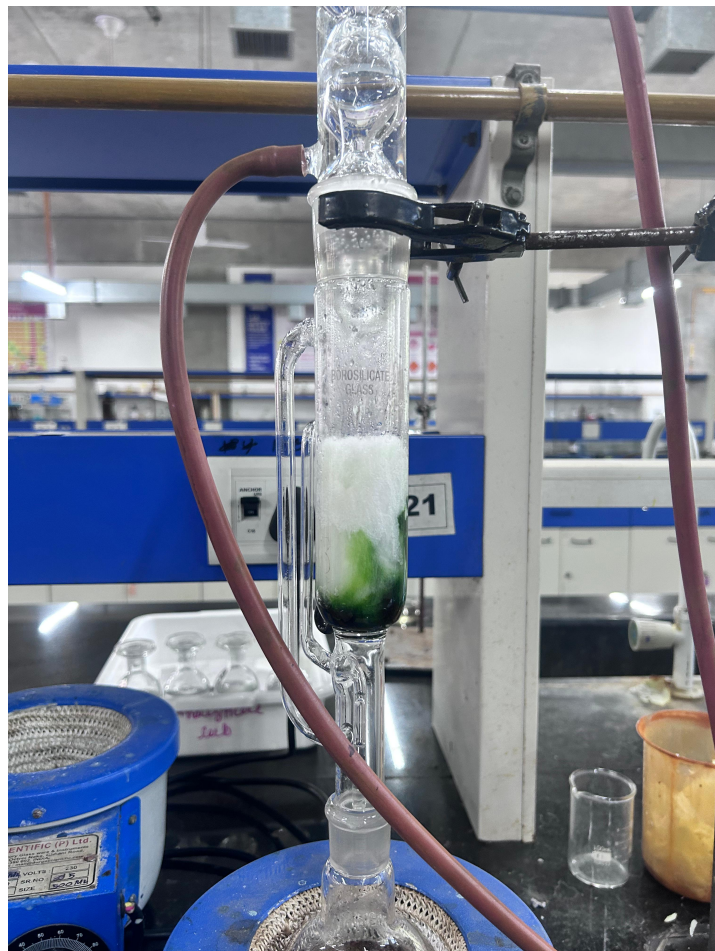
- Microbial culture spread on agar plates
- Extract applied using disc / well diffusion method
- Plates incubated at 37°C for 24 hours

Observation

- Zone of inhibition measured
- Larger zone indicates higher antimicrobial activity

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Prototype Image/Video



Soxhlet Method



Rotary Vacuum Evaporator



Microbial Activity

Learning Outcome

Technical/Conceptual Learning- Gained plant based extraction methods and about antimicrobial activity testing

Problem-Solving and Tinkering Skills Developed- Improved ability to fix errors and handle laboratory instruments

Learning from Mentoring Sessions and Feedback- Developed confidence through learning to identify mistakes and make necessary corrections under proper guidance

Teamwork and Collaboration- Improved coordination and problem-solving through proper communication

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Conclusion & Future Scope

Conclusion- Neem leaf extract showed noticeable antimicrobial activity, indicating its potential as a natural biopesticide.

Summary of Achievements- Successfully extracted neem compounds using Soxhlet method and tested their activity on microbes

Future Enhancements and Applications- Further purification and testing can improve effectiveness and can be used for agricultural pest control.

Reflection on the Project Experience- This project enhanced our laboratory skills, improved managing our team coordination and understanding of extraction techniques

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Do you want to apply for PoC/Prototype Development support under Students Startup and Innovation Policy (SSIP 2.0) under GUIITAR Council?

We are currently not applying for PoC/Prototype Development support under SSIP 2.0.

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➤➤ **Thank You..**